

SECURITY COMPLIANCE DECLARATION & SOVEREIGN ATTESTATION

CONNECTING EUROPE FACILITY (CEF) DIGITAL // SMART CABLES WORKS

- **Project Acronym:** AETERNA-SCW
 - **Proposal ID:** CEF-DIG-2026-SMART-CABLES-101354145
 - **Applicant Lead:** AETERNA (POMORIE, BULGARIA)
 - **Participant Identification Code (PIC):** 865986222
 - **Infrastructure Target:** Black Sea & Eastern Mediterranean Submarine Optical Telecommunications Trunks
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DECLARATION OF COMPLIANCE WITH EU SECURITY REQUIREMENT & CRITICAL INFRASTRUCTURE SOVEREIGNTY

I, the undersigned, acting as the Sovereign Systems Architect and authorized legal representative of **AETERNA**, hereby declare and certify the following security compliance metrics regarding the deployment and operation of the **AETERNA-SCW (AIGIS Subsea Shield)** system:

1. Zero Cloud Dependency & Data Sovereignty

The AETERNA-SCW control plane operates under a strict **Zero-Entropy, On-Premise Execution** paradigm. There is no transmission of telemetry, packet headers, metadata, or phase shift statistics to third-party or non-EU cloud service providers. All Distributed Acoustic Sensing (DAS) and State of Polarization (SOP) monitoring data is processed locally at submarine cable landing-station edge nodes within the sovereign territory of the European Union.

2. Clean-Room Software Stack & Zero Untrusted Third-Country Technology

All core mathematical layers, including the real-time subsea signal classifiers, are built using a native, vectorized codebase (compiled via the **Mojo** system with integer-only math). We guarantee that: * No software libraries or hardware components in the hot execution path originate from jurisdictions outside of the European Union or NATO member states. * No closed-source, proprietary components from third countries are utilized, bypassing logical

backdoors. * The system is compliant with the **EU 5G Toolbox** and critical telecommunications infrastructure supply-chain guidelines.

3. Active Cyber-Physical Defense & eBPF Process Isolation

Landing-station SCADA terminals are hardened using kernel-level **eBPF (Extended Berkeley Packet Filter)** **Sentinel** containment modules. In the event of physical fiber tampering or line tapping, AIGIS triggers instant process apoptosis and trunk traffic isolation in **under 1.02ms**, preventing data extraction without relying on traditional network routing layers which are vulnerable to lateral exploits.

4. Compliance with EU Regulatory Frameworks

The proposed system architectures and organizational controls of AETERNA are fully compliant with: * The **NIS2 Directive** (Directive EU 2022/2555) on cyber security risk-management measures. * The **General Data Protection Regulation (GDPR)**, as all patient, consumer, and transit telecommunication data is completely isolated and never stored or inspected. * The **EU AI Act**, certifying that our real-time Mojo classifiers operate as deterministic mathematical signal separators and do not engage in biometric tracking or high-risk profiling.

SIGNATURE & ATTESTATION

Declared by: AETERNA Neural QA Nexus

Authorizing Official: DIMITAR PRODROMOV (Sovereign Systems Architect)

Entity: AETERNA (POMORIE, BG)

Date: 2026-05-20

Signature Applied Electronically under PIC **865986222**

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